Software Engineering & Project Management (SEPM)

Energy & Utility Management System (EUMS)

Electricity · Water · Gas — billing, metering and customer service

UML Diagrams

Software process model: Waterfall · Combined SEPM + DBMS course project

Group number	Group __
Course and section	Software Engineering & Project Management (SEPM), Section __
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Date	October 2026

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1. UML Diagrams

1.1 Sequence diagram – meter reading to bill to payment

This is the main end-to-end flow and the one shown in the demo video.

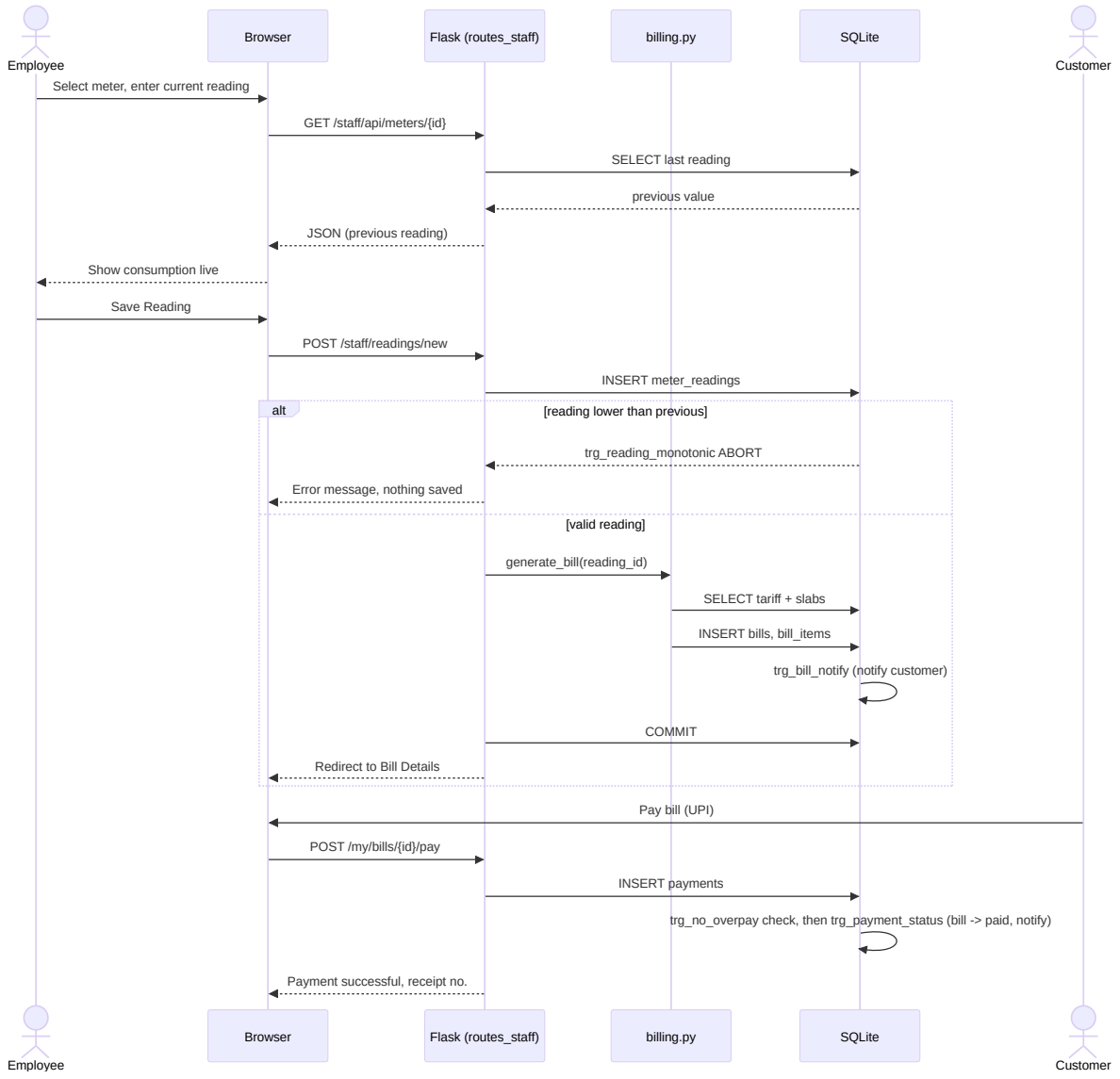


Figure 1: Sequence diagram: record reading, generate bill, pay bill

1.2 Activity diagram – online bill payment

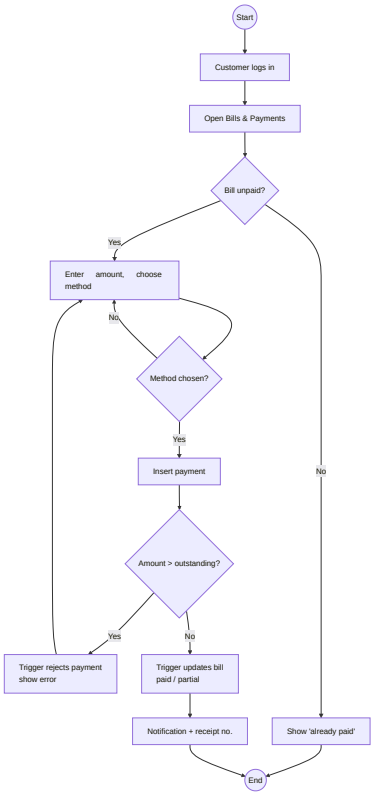


Figure 2: Activity diagram for paying a bill

1.3 Class diagram (domain model)

The application is written in a procedural Flask style, so these classes are the *domain model*: each class is a database table, and the operations are carried out by route functions and `billing.py`.

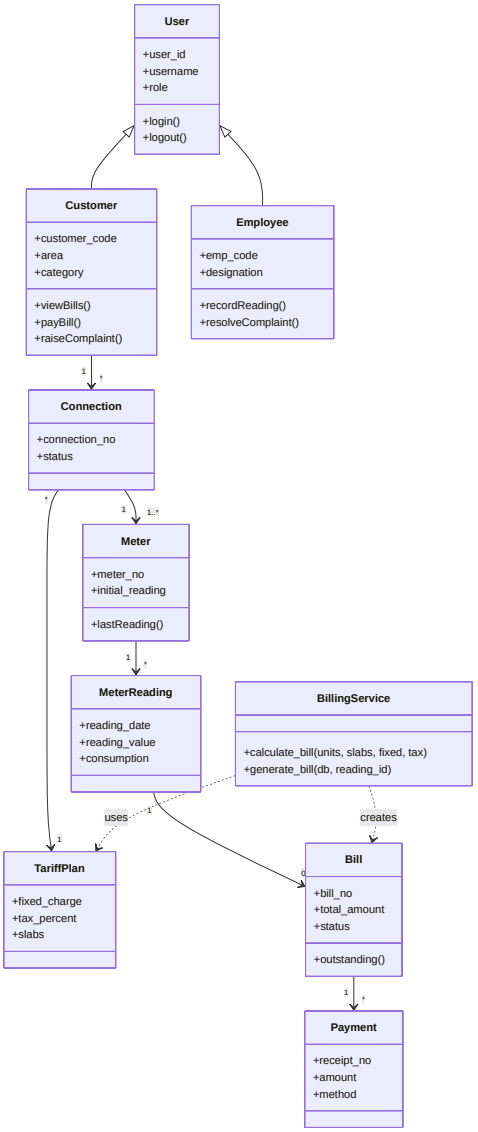


Figure 3: Class diagram of the main domain entities